



Lunar Cloud Platform 2.6

# Electronic Health Information Export



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# Lunar EHI Export Overview

Lunar's EHI Export functionality enables healthcare organizations to extract Electronic Health Information (EHI) from the Lunar Cloud Platform in compliance with §170.315(b)(10) of the ONC 2015 Edition Cures Update. The platform supports single-patient and bulk-patient data extraction in the form of an "export zip" file containing machine-readable files, enabling authorized users to access and manage the designated record set as defined under 45 CFR 164.502.

The Lunar Cloud Platform leverages the C-CDA (Consolidated Clinical Document Architecture) format for the majority of patient EHI, adhering to widely recognized industry standards for health data interoperability. The C-CDA XML files provide structured and comprehensive representations of clinical information, facilitating the smooth transfer and usability of EHI across systems.

In addition to the C-CDA format, certain EHI data - such as images, scanned documents, and multimedia files - are included as comma-separated values (csv) files. These supplemental files ensure a complete and cohesive representation of the patient's health information.

The content and format of the exported EHI may vary based on:

- **Lunar Cloud Platform Configuration:** Organization-specific settings and workflows.
- **Implementation Details:** Standard implementations include the Lunar Clinical Suite, Lunar Revenue Suite, Lunar Lab Suite, Lunar IT Suite, Lunar Operations Suite, Lunar Analytics Suite, and Lunar Patient Suite.
- **Software Version:** The current version of the Lunar Cloud Platform in use.
- **Organizational Practices:** Data documentation, input workflows, and system usage.

# C-CDA Format Guide

The Consolidated Clinical Document Architecture (C-CDA) is a standardized format developed by HL7 (Health Level Seven International) to enable the exchange of structured clinical documents across different health systems. It serves as the foundation for the majority of patient data exported from the Lunar Cloud Platform.

C-CDA is an XML-based format that organizes and represents patient health information in a machine-readable, interoperable structure. It provides a standardized way to store and share critical health data, ensuring consistency and compatibility between EHR systems.

The Lunar Cloud Platform uses CDA Release 4.1. The official HL7 documentation for C-CDA can be found at the following URL:

[https://www.hl7.org/implement/standards/product\\_brief.cfm?product\\_id=447](https://www.hl7.org/implement/standards/product_brief.cfm?product_id=447)

# C-CDA Document Sections

The following sections outline the core components of a C-CDA document from the Lunar Cloud Platform, each representing a specific aspect of a patient's health record. These sections are designed to provide a clear and organized summary of clinical information, including encounters, medications, procedures, and more.

Additionally, the last section, Supplemental Data, is a proprietary Lunar data format that supplements a patient's C-CDA with any additional electronic health information in a series of comma-separated values (csv) files that's not captured by the standard C-CDA format.

## Patient Summary

This section includes basic demographic information about the patient, such as name, date of birth, gender, contact details, and identifiers (e.g., MRN). It serves as the foundational metadata for identifying the patient within the document.

### Example XML

The following example sits within a `<recordTarget></recordTarget>` set of tags.

```
<patientRole>
  <id root="1.3.6.1.4.1.38630.2.1.1.117" extension="M02389491"/>
  <id root="2.16.840.1.113883.4.1" extension="888-88-8888"/>
  <addr>
    <streetAddressLine>3 Lunar Dr</streetAddressLine>
    <city>Moon Township</city>
    <state>CA</state>
    <postalCode>94107</postalCode>
    <country>USA</country>
  </addr>
  <telecom value="tel:+12062702651"/>
  <patient>
    <name>
      <given>Walter</given>
      <family>Boron</family>
    </name>
    <administrativeGenderCode code="M" displayName="Male" codeSystem="2.16.840.1.113883.5.1"
codeSystemName="AdministrativeGender"/>
    <birthTime value="19321014"/>
    <maritalStatusCode code="M" displayName="Married" codeSystem="2.16.840.1.113883.5.2"/>
    <raceCode code="2106-3" displayName="White" codeSystem="2.16.840.1.113883.6.238"
codeSystemName="CDC - Race and Ethnicity"/>
    <ethnicGroupCode displayName="Not Hispanic or Latino" code="2186-5"
codeSystem="2.16.840.1.113883.6.238" codeSystemName="Race and Ethnicity - CDC"/>
    <languageCommunication>
      <languageCode code="en"/>
    </languageCommunication>
  </patient>
  <providerOrganization>
    <id root="10e46398-fead-45d2-b2bb-53d769cde8c1"/>
    <name>Lunar Systems</name>
    <telecom value="tel:+18007654321" use="WP"/>
    <addr>
      <streetAddressLine>1 Lunar Dr</streetAddressLine>
      <city>Moon Township</city>
      <state>CA</state>
```

```

    <postalCode>94107</postalCode>
    <country>USA</country>
  </addr>
</providerOrganization>
</patientRole>

```

## Key elements

| Element                    | Description                                                                                 |
|----------------------------|---------------------------------------------------------------------------------------------|
| <patientRole>              | Wraps the information about the patient                                                     |
| <id>                       | Captures unique patient identifiers (e.g., MRN or SSN)                                      |
| <addr>                     | Specifies the patient's address (e.g., street, city, state, postal code, and country)       |
| <telecom>                  | Contains the patient's contact number or email                                              |
| <patient>                  | Wraps patient-specific details such as name, gender, birth date, and ethnicity              |
| <name>                     | Contains the patient's name, with separate elements for given and family names              |
| <administrativeGenderCode> | Indicates the patient's gender                                                              |
| <birthTime>                | Specifies the patient's date of birth in YYYYMMDD format                                    |
| <maritalStatusCode>        | Represents the marital status using a coded value (e.g., Married - M)                       |
| <raceCode>                 | Describes the patient's race and ethnicity according to CDC standards                       |
| <ethnicGroupCode>          | Describes the patient's race and ethnicity according to CDC standards                       |
| <languageCommunication>    | Captures the patient's preferred language for communication (e.g., en-US for English)       |
| <providerOrganization>     | Contains the associated healthcare organization details, such as name, address, and contact |
| <deceasedInd>              | Indicates if the patient is deceased.                                                       |
| <deceasedTime>             | Specifies the patient's date of death in YYYYMMDD format, if deceased                       |

## Allergies and Intolerances

The Allergies and Intolerances section lists any known allergies or adverse reactions the patient has experienced to medications, substances, or environmental factors. Each entry includes the allergen, reaction type, and severity, helping to prevent adverse clinical events.

### Example XML

The following example sits within an allergies `<section></section>` set of tags.

#### Metadata

This defines the section metadata such as the template ID and LOINC code.

```
<templateId root="2.16.840.1.113883.10.20.22.2.6.1" extension="2015-08-01"/>
<code code="48765-2" codeSystem="2.16.840.1.113883.6.1" codeSystemName="LOINC"
displayName="Allergies and adverse reactions Document"/>
<title>ALLERGIES, ADVERSE REACTIONS, ALERTS</title>
```

#### Human-Readable Text

This provides the narrative representation of the allergies in a table format.

```
<text>
<table border="1" width="100%">
  <thead>
    <tr>
      <th>Substance</th>
      <th>Reaction</th>
      <th>Severity</th>
      <th>Status</th>
    </tr>
  </thead>
  <tbody>
    <tr>
      <td><content>Latex</content></td>
      <td><content>Rash</content></td>
      <td><content>Moderate</content></td>
      <td><content>Active</content></td>
    </tr>
  </tbody>
</table>
</text>
```



## Coded Entries

This contains the structured data for each allergy or adverse reaction. `<entry>` contains details about the allergy, including substance, severity, and the effective time.

```
<entry typeCode="DRIV">
  <observation classCode="OBS" moodCode="EVN">
    <templateId root="2.16.840.1.113883.10.20.22.4.7"/>
    <id root="12345" extension="Allergy1"/>
    <code code="ASSERTION" codeSystem="2.16.840.1.113883.5.4"/>
    <statusCode code="completed"/>
    <effectiveTime>
      <low value="20241201"/>
    </effectiveTime>
    <value displayName="Propensity to adverse reaction to drug" code="419511003" xsi:type="CD"
codeSystem="2.16.840.1.113883.6.96" codeSystemName="SNOMED CT"/>
    <participant typeCode="CSM">
      <participantRole classCode="MANU">
        <playingEntity classCode="MMAT">
          <code code="7980" codeSystem="2.16.840.1.113883.6.88" codeSystemName="RxNorm"
displayName="Penicillin G"/>
        </playingEntity>
      </participantRole>
    </participant>
    <entryRelationship typeCode="MFST">
      <observation classCode="OBS" moodCode="EVN">
        <code code="271807003" displayName="Rash" codeSystem="2.16.840.1.113883.6.96"
codeSystemName="SNOMED-CT"/>
        <value xsi:type="CD" code="6736007" displayName="Moderate" codeSystem="2.16.840.1.113883.6.96"/>
      </observation>
    </entryRelationship>
  </observation>
</entry>
```

## Key elements in Coded Entries

| Element         | Description                                      |
|-----------------|--------------------------------------------------|
| <entry>         | Represents the allergy or adverse reaction entry |
| <observation>   | Contains details about the allergy               |
| <statusCode>    | Indicates the status (e.g., active, completed)   |
| <effectiveTime> | Specifies the time of occurrence/documentation   |
| <code>          | Coded representation of the allergen or reaction |

|                     |                                               |
|---------------------|-----------------------------------------------|
| <value>             | Coded representation of the reaction severity |
| <entryRelationship> | Links related observations, such as severity  |

## Problem List

This section contains the patient's active and resolved medical problems, diagnoses, and conditions. Problems are documented with descriptions, diagnosis codes, status, and dates to provide a summary of ongoing and historical health issues.

### Example XML

The following example sits within a problem list `<section></section>` set of tags.

#### Metadata

This defines the section metadata such as the template ID and LOINC code.

```
<templateId root="2.16.840.1.113883.10.20.22.2.5.1" extension="2015-08-01"/>
<code code="11450-4" codeSystem="2.16.840.1.113883.6.1" displayName="Problem list"/>
<title>PROBLEMS</title>
```

#### Human-Readable Text

This provides the narrative representation of the problem list in a table format.

```
<text>
<table border="1" width="100%">
  <thead>
    <tr>
      <th>Problem</th>
      <th>Start Date</th>
      <th>End Date</th>
      <th>Status</th>
    </tr>
  </thead>
  <tbody>
    <tr>
      <td>Acute exacerbation of COPD</td>
      <td>2023-12-01</td>
      <td></td>
      <td>Active</td>
    </tr>
  </tbody>
</table>
</text>
```

## Coded Entries

This contains the structured data for each problem in the problem list. Each `<entry>` corresponds to a specific clinical concern.

```
<entry typeCode="DRIV">
  <act classCode="ACT" moodCode="EVN">
    <templateId root="2.16.840.1.113883.10.20.22.4.3" extension="2015-08-01"/>
    <id root="12345" extension="Problem1"/>
    <code code="CONC" displayName="Concern" codeSystem="2.16.840.1.113883.5.6"
codeSystemName="ACTCLASS"/>
    <statusCode code="active"/>
    <effectiveTime>
      <low value="20231201"/>
    </effectiveTime>
    <entryRelationship typeCode="SUBJ">
      <observation classCode="OBS" moodCode="EVN">
        <templateId root="2.16.840.1.113883.10.20.22.4.4" extension="2015-08-01"/>
        <id root="12345" extension="Observation1"/>
        <code code="409586006" displayName="Complaint" codeSystem="2.16.840.1.113883.6.96"
codeSystemName="SNOMED-CT">
          <translation displayName="Complaint" code="75322-8" codeSystem="2.16.840.1.113883.6.1"
codeSystemName="LOINC"/>
        </code>
        <statusCode code="completed"/>
        <effectiveTime>
          <low value="20231201"/>
        </effectiveTime>
        <value xsi:type="CD" code="257004" displayName="Acute exacerbation of COPD"
codeSystem="2.16.840.1.113883.6.96" codeSystemName="SNOMED CT"/>
        </observation>
      </entryRelationship>
    </act>
  </entry>
```

## Key elements in Coded Entries

| Element             | Description                                             |
|---------------------|---------------------------------------------------------|
| <entry>             | Represents a problem or concern entry                   |
| <act>               | Serves as a wrapper for the Problem Concern             |
| <code>              | Describes the concern or problem type                   |
| <statusCode>        | Indicates the status (e.g., active, completed)          |
| <effectiveTime>     | Specifies the timeframe for the problem                 |
| <statusCode>        | Indicates the status (e.g., active, completed)          |
| <entryRelationship> | Links the concern act to the problem observation        |
| <observation>       | Captures the detailed clinical problem or diagnosis     |
| <value>             | Describes the problem using standardized coding systems |



## History of Medication Use

This section contains information about a patient's historical medication usage, including the medication name, dosage, frequency, route of administration, and status.

### Example XML

The following example sits within a history of medication use `<section></section>` set of tags.

#### Metadata

This defines the section metadata, such as the template ID and LOINC code.

```
<templateId root="2.16.840.1.113883.10.20.22.2.1.1" extension="2014-06-09"/>
<code code="10160-0" codeSystem="2.16.840.1.113883.6.1" displayName="HISTORY OF MEDICATION USE"/>
<title>HISTORY OF MEDICATION USE</title>
```

#### Human-Readable Text

This provides the narrative representation of the history of medication use in a table format.

```
<text>
<table border="1" width="100%">
  <thead>
    <tr>
      <th>Medication</th>
      <th>Directions</th>
      <th>Status</th>
    </tr>
  </thead>
  <tbody>
    <tr>
      <td>aspirin</td>
      <td>PO, 81 mg, qday, REG</td>
      <td>Active</td>
    </tr>
    <tr>
      <td>bisoprolol fumarate</td>
      <td>PO, 5 mg, qday, REG</td>
      <td>Completed</td>
    </tr>
  </tbody>
</table>
</text>
```

## Coded Entries

This contains the structured data for each medication entry. Each `<entry>` corresponds to a specific medication administration or order.

```
<entry typeCode="DRIV">
  <substanceAdministration classCode="SBADM" moodCode="EVN" negationInd="false">
    <templateId root="2.16.840.1.113883.10.20.22.4.16" extension="2014-06-09"/>
    <id root="a1b2c3d4-e567-89f0-1234-56789abcdef0" />
    <statusCode code="completed"/>
    <effectiveTime xsi:type="IVL_TS">
      <low value="20231201"/>
    </effectiveTime>
    <doseQuantity value="81" unit="mg"/>
    <rateQuantity value="1" unit="day"/>
    <routeCode code="C38288" displayName="Oral route" codeSystem="2.16.840.1.113883.3.26.1.1"
codeSystemName="NCI Thesaurus"/>
    <consumable>
      <manufacturedProduct classCode="MANU">
        <templateId root="2.16.840.1.113883.10.20.22.4.23" extension="2014-06-09"/>
        <manufacturedMaterial>
          <code code="1191" displayName="aspirin" codeSystem="2.16.840.1.113883.6.88"
codeSystemName="RxNorm"/>
        </manufacturedMaterial>
      </manufacturedProduct>
    </consumable>
  </substanceAdministration>
</entry>
```

## Key elements in Coded Entries

| Element                   | Description                                                        |
|---------------------------|--------------------------------------------------------------------|
| <entry>                   | Represents an individual medication administration record          |
| <substanceAdministration> | Captures the structured details of the medication use              |
| <statusCode>              | Indicates if the medication is active or completed                 |
| <effectiveTime>           | Start and end date of the medication use                           |
| <doseQuantity>            | Specifies the dosage amount for the medication                     |
| <rateQuantity>            | Indicates the rate of administration, if applicable                |
| <routeCode>               | Specifies the method of administration (e.g., oral, IV)            |
| <consumable>              | Links the medication to a recognized code (e.g., RxNorm)           |
| <manufacturedProduct>     | Represents the medication product details                          |
| <manufacturedMaterial>    | Contains the code and name of the medication (e.g., CVX or RxNorm) |

## Laboratory/Diagnostic Results

The Laboratory Results section provides lab test results, including names of tests, values, units of measurement, reference ranges, and interpretation (e.g., abnormal/normal).

### Example XML

The following example sits within a history of medication use `<section></section>` set of tags.

#### Metadata

This defines the section metadata, such as the template ID and LOINC code.

```
<templateId root="2.16.840.1.113883.10.20.22.2.3.1" extension="2015-08-01"/>
<code code="30954-2" codeSystem="2.16.840.1.113883.6.1" displayName="Diagnostic / Lab Results"/>
<title>DIAGNOSTIC IMAGING / LAB RESULTS</title>
```

#### Human-Readable Text

This provides the narrative representation of laboratory results in a table format.

```
<text>
<table border="1" width="100%">
  <thead>
    <tr>
      <th>Result Type</th>
      <th>Result Value</th>
      <th>Date/Time</th>
    </tr>
  </thead>
  <tbody>
    <tr>
      <td>wbc_white_blood_cells</td>
      <td>8.54</td>
      <td>2024-12-16 20:08:28Z</td>
    </tr>
    <tr>
      <td>mchc</td>
      <td>35 g/dL</td>
      <td>2024-12-16 20:08:28Z</td>
    </tr>
    <tr>
      <td>hb_hemoglobin</td>
      <td>15 g/dL</td>
      <td>2024-12-16 20:08:28Z</td>
    </tr>
  </tbody>
</table>
```

```

</tr>

</tbody>
</table>
</text>

```

## Coded Entries

This contains the structured data for each laboratory or diagnostic result. Each `<entry>` corresponds to a specific laboratory result.

```

<entry typeCode="DRIV">
  <organizer classCode="BATTERY" moodCode="EVN">
    <templateId root="2.16.840.1.113883.10.20.22.4.1" extension="2015-08-01"/>
    <id root="c5281500-46d6-497e-876a-01ed5a1aa3a1"/>
    <code code="58410-2" displayName="cbc_auto_complete" codeSystem="2.16.840.1.113883.6.1"
codeSystemName="LOINC"/>
    <statusCode code="completed"/>
    <effectiveTime>
      <low value="20241216200828+0000"/>
      <high value="20241216200828+0000"/>
    </effectiveTime>
    <!-- White Blood Cell Count -->
    <component>
      <observation classCode="OBS" moodCode="EVN">
        <templateId root="2.16.840.1.113883.10.20.22.4.2" extension="2015-08-01"/>
        <id root="5ec89403-80da-4f6f-8808-6497f6e1fc2f"/>
        <code code="6690-2" displayName="wbc_white_blood_cells" codeSystem="2.16.840.1.113883.6.1"/>
        <statusCode code="completed"/>
        <effectiveTime value="20241216200828+0000"/>
        <value xsi:type="PQ" value="8.54" unit="10*3/uL"/>
      </observation>
    </component>
    <!-- MCHC -->
    <component>
      <observation classCode="OBS" moodCode="EVN">
        <templateId root="2.16.840.1.113883.10.20.22.4.2" extension="2015-08-01"/>
        <id root="25a3e47e-e23f-440b-96ed-75b2098d3a85"/>
        <code code="786-4" displayName="mchc" codeSystem="2.16.840.1.113883.6.1"/>
        <statusCode code="completed"/>
        <effectiveTime value="20241216200828+0000"/>
        <value xsi:type="PQ" value="35" unit="g/dL"/>
      </observation>
    </component>
    <!-- Hemoglobin -->
    <component>
      <observation classCode="OBS" moodCode="EVN">
        <templateId root="2.16.840.1.113883.10.20.22.4.2" extension="2015-08-01"/>

```

```

<id root="69194e53-22d1-479c-832d-88f4e11dd3ea"/>
<code code="718-7" displayName="hb_hemoglobin" codeSystem="2.16.840.1.113883.6.1"/>
<statusCode code="completed"/>
<effectiveTime value="20241216200828+0000"/>
<value xsi:type="PQ" value="15" unit="g/dL"/>
</observation>
</component>
</organizer>
</entry>

```

## Key elements in Coded Entries

| Element         | Description                                                            |
|-----------------|------------------------------------------------------------------------|
| <entry>         | Represents a laboratory or diagnostic result entry                     |
| <organizer>     | Groups related observations (e.g., a lab panel or test set)            |
| <observation>   | Represents each individual test result                                 |
| <code>          | Identifies the test or panel (e.g., 6690-2 for white blood cell count) |
| <value>         | Contains the result value and unit of measurement                      |
| <effectiveTime> | Indicates the date/time the test was performed                         |
| <statusCode>    | Indicates the status of the result (e.g., "completed")                 |

## Procedures

The Procedures section details medical or surgical procedures the patient has undergone. It includes procedure names, dates, descriptions, and providers involved, ensuring a record of interventions is available for continuity of care.

### Example XML

The following example sits within a procedures `<section></section>` set of tags.

#### Metadata

This defines the section metadata, such as the template ID and LOINC code.

```
<templateId root="2.16.840.1.113883.10.20.22.2.7.1" extension="2014-06-09"/>
<code code="47519-4" codeSystem="2.16.840.1.113883.6.1" codeSystemName="LOINC"
displayName="HISTORY OF PROCEDURES"/>
<title>PROCEDURES SECTION</title>
```

#### Human-Readable Text

This provides the narrative representation of procedures in a table format.

```
<text>
<table border="1" width="100%">
  <thead><tr>
    <th>Procedure</th>
    <th>Date</th>
    <th>Details</th>
  </tr></thead>
  <tbody><tr>
    <td>Total Knee Replacement (Left)</td>
    <td>2006-03-08</td>
    <td></td>
  </tr><tr>
    <td>Colonoscopy - nil of note</td>
    <td>2005-05-18</td>
    <td></td>
  </tr><tr>
    <td>Excision - Benign lesion left forehead</td>
    <td>2003-09-05</td>
    <td></td>
  </tr>
</tbody>
</table>
</text>
```



## Coded Entries

This contains the structured data for each procedure. Each `<entry>` corresponds to a specific procedure.

```
<entry typeCode="DRIV">
  <procedure classCode="PROC" moodCode="EVN">
    <templateId root="2.16.840.1.113883.10.20.22.4.14"/>
    <id root="522b0283-eea1-4502-b05c-28a1cd953fbf"/>
    <code displayName="Total Knee Replacement (Left)"
      code="443681002" codeSystem="2.16.840.1.113883.6.96"
      codeSystemName="SNOMED-CT"/>
    <statusCode code="completed"/>
    <effectiveTime value="20060308"/>
    <author>
      <assignedAuthor>
        <id root="2.16.840.1.113883.4.6" extension="1234567890"/>
        <assignedPerson>
          <name>
            <given>John</given><family>Smith</family>
          </name>
        </assignedPerson>
      </assignedAuthor>
    </author>
  </procedure>
</entry>
```

### Key elements in Coded Entries

| Element         | Description                                                     |
|-----------------|-----------------------------------------------------------------|
| <entry>         | Represents an individual procedure record                       |
| <procedure>     | Captures structured details of the procedure                    |
| <code>          | Describes the specific procedure (e.g., SNOMED-CT code)         |
| <statusCode>    | Indicates the status of the procedure (e.g., completed)         |
| <effectiveTime> | Specifies the date or time of the procedure                     |
| <author>        | Identifies the person responsible for documenting the procedure |

## Social History

This section includes details about the patient's lifestyle and social factors that influence health, such as smoking status, alcohol use, employment, and living arrangements.

### Example XML

The following example sits within a social history `<section></section>` set of tags.

#### Metadata

This defines the section metadata, such as the template ID and LOINC code.

```
<templateId root="2.16.840.1.113883.10.20.22.2.17" extension="2015-08-01"/>
<code code="29762-2" codeSystem="2.16.840.1.113883.6.1" displayName="Social History"/>
<title>SOCIAL HISTORY</title>
```

#### Human-Readable Text

This provides the narrative representation of social history in a table format.

```
<text>
<table border="1" width="100%">
  <thead>
    <tr>
      <th>Social History Observation</th>
      <th>Description</th>
      <th>Dates Observed</th>
    </tr>
  </thead>
  <tbody>
    <tr>
      <td>Tribal Affiliation</td>
      <td>Coquille Indian Tribe</td>
      <td>20150622</td>
    </tr>
  </tbody>
</table>
</text>
```

## Coded Entries

This contains the structured data for social history. Each `<entry>` corresponds to a specific social history observation.

```
<entry typeCode="DRIV">
  <observation classCode="OBS" moodCode="EVN">
    <templateId root="2.16.840.1.113883.10.20.22.4.506" extension="2023-05-01"/>
    <id root="2fa90a26-4f6b-4764-b555-fd20d64b2346"/>
    <code code="95370-3" codeSystem="2.16.840.1.113883.6.1" codeSystemName="LOINC"
      displayName="Tribal affiliation"/>
    <statusCode code="completed"/>
    <effectiveTime value="20150622"/>
    <value xsi:type="CD" code="65" displayName="Coquille Indian Tribe"
      codeSystem="2.16.840.1.113883.5.140" codeSystemName="TribalEntityUS"/>
  </observation>
</entry>
```

### Key elements in Coded Entries

| Element                          | Description                                                                                                     |
|----------------------------------|-----------------------------------------------------------------------------------------------------------------|
| <code>&lt;entry&gt;</code>       | Represents an individual social history observation                                                             |
| <code>&lt;observation&gt;</code> | Captures structured details of the social history event                                                         |
| <code>&lt;code&gt;</code>        | Describes the specific type of social history (e.g., smoking)                                                   |
| <code>&lt;statusCode&gt;</code>  | Indicates the status of the observation (e.g., completed)                                                       |
| <code>&lt;value&gt;</code>       | Captures the value of the observation (e.g., ex-smoker) and links to a standardized code (e.g., SNOMED-CT code) |

## Functional Status

This section documents the patient's functional and cognitive abilities, including mobility, activities of daily living, and any physical or mental impairments.

### Example XML

The following example sits within a functional status `<section></section>` set of tags.

#### Metadata

This defines the section metadata, such as the template ID and LOINC code.

```
<templateId root="2.16.840.1.113883.10.20.22.2.14" extension="2014-06-09"/>
<code code="47420-5" codeSystem="2.16.840.1.113883.6.1" codeSystemName="LOINC"
displayName="Functional Status"/>
<title>FUNCTIONAL STATUS</title>
```

#### Human-Readable Text

This provides the narrative representation of functional status in a table format.

```
<text>
<table border="1" width="100%">
  <thead>
    <tr>
      <th>Functional Category</th>
      <th>Functional Status</th>
      <th>Problem</th>
      <th>Effective Dates</th>
    </tr>
  </thead>
  <tbody>
    <tr>
      <td ID="FS_a97eb25a-afc3-488f-a259-119f5977b2b1__0">Dependence on Cane</td>
      <td/>
      <td>2005-05-01 - </td>
      <td/>
    </tr>
  </tbody>
</table>
</text>
```

## Coded Entries

This contains the structured data for functional status. Each `<entry>` corresponds to a specific functional status observation.

```
<entry typeCode="DRIV">
  <observation classCode="OBS" moodCode="EVN">
    <templateId root="2.16.840.1.113883.10.20.22.4.67" extension="2014-06-09"/>
    <id root="a1b2c3d4-e567-8901-2345-6789abcdef01"/>
    <code code="371150009" displayName="Walking disability"
      codeSystem="2.16.840.1.113883.6.96" codeSystemName="SNOMED CT"/>
    <statusCode code="completed"/>
    <effectiveTime>
      <low value="20050501"/>
    </effectiveTime>
    <value xsi:type="CD" code="105504002" displayName="Dependence on Cane"
      codeSystem="2.16.840.1.113883.6.96" codeSystemName="SNOMED CT"/>
  </observation>
</entry>
```

### Key elements in Coded Entries

| Element                            | Description                                                                                                            |
|------------------------------------|------------------------------------------------------------------------------------------------------------------------|
| <code>&lt;entry&gt;</code>         | Represents an individual functional status observation                                                                 |
| <code>&lt;observation&gt;</code>   | Captures structured details of the functional status event                                                             |
| <code>&lt;code&gt;</code>          | Describes the functional status (e.g., walking disability)                                                             |
| <code>&lt;statusCode&gt;</code>    | Indicates the status of the observation (e.g., completed)                                                              |
| <code>&lt;effectiveTime&gt;</code> | Defines the period when the observation applies                                                                        |
| <code>&lt;value&gt;</code>         | Captures the value of the observation (e.g., needs assistance) and links to a standardized code (e.g., SNOMED-CT code) |

## Mental/Cognitive Status

The Mental/Cognitive Status Section documents observations related to a patient's mental and cognitive health, such as assessments for conditions like dementia, depression, or mental status evaluations.

### Example XML

The following example sits within a mental/cognitive status `<section></section>` set of tags.

#### Metadata

This defines the section metadata, such as the template ID and LOINC code.

```
<templateId root="2.16.840.1.113883.10.20.22.2.56" extension="2015-08-01"/>
<code code="10190-7" codeSystem="2.16.840.1.113883.6.1" codeSystemName="LOINC"
displayName="Mental/Cognitive Status"/>
<title>MENTAL/COGNITIVE STATUS</title>
```

#### Human-Readable Text

This provides the narrative representation of functional status in a table format.

```
<text>No Mental Status Information.</text>
```

#### Coded Entries

This contains the structured data for mental/cognitive status. Each `<entry>` corresponds to a specific mental or cognitive status observation.

```
<entry typeCode="DRIV">
  <observation classCode="OBS" moodCode="EVN">
    <templateId root="2.16.840.1.113883.10.20.22.4.74" extension="2015-08-01"/>
    <id root="d8c31e1b-8b3d-4e3b-a9e2-7ef6d53d9b1e"/>
    <code code="386924004" displayName="Mental status assessment"
      codeSystem="2.16.840.1.113883.6.96" codeSystemName="SNOMED CT"/>
    <statusCode code="completed"/>
    <effectiveTime>
      <low value="20231201"/>
    </effectiveTime>
    <value xsi:type="CD" code="248233002" displayName="No cognitive impairment detected"
      codeSystem="2.16.840.1.113883.6.96" codeSystemName="SNOMED CT"/>
  </observation>
</entry>
```

## Key elements in Coded Entries

| Element         | Description                                                                                                          |
|-----------------|----------------------------------------------------------------------------------------------------------------------|
| <entry>         | Represents an individual mental/cognitive status observation                                                         |
| <observation>   | Captures structured details of the cognitive status assessment                                                       |
| <code>          | Describes the mental/cognitive assessment and links to a standardized code (e.g., SNOMED-CT code)                    |
| <statusCode>    | Indicates the status of the observation (e.g., completed)                                                            |
| <effectiveTime> | Defines the period when the observation applies                                                                      |
| <value>         | Captures the outcome of the assessment (e.g., no impairment) and links to a standardized code (e.g., SNOMED-CT code) |

## Vital Signs

The Vital Signs section records key physiological measurements such as blood pressure, heart rate, temperature, respiratory rate, height, weight, and BMI.

### Example XML

The following example sits within a vital signs `<section></section>` set of tags.

#### Metadata

This defines the section metadata, such as the template ID and LOINC code.

```
<templateId root="2.16.840.1.113883.10.20.22.2.4.1" extension="2015-08-01"/>
<code code="8716-3" codeSystem="2.16.840.1.113883.6.1" codeSystemName="LOINC" displayName="Vital Signs"/>
<title>VITAL SIGNS</title>
```

#### Human-Readable Text

This provides the narrative representation of vital signs in a table format.

```
<text>
<table border="1" width="100%">
  <thead>
    <tr>
      <th>Name</th>
      <th>Value</th>
      <th>Date/Time</th>
    </tr>
  </thead>
  <tbody>
    <tr>
      <td>o2_sats</td>
      <td>92 %</td>
      <td>2024-11-24 04:30:00Z</td>
    </tr>
    <tr>
      <td>temp</td>
      <td>38 Cel</td>
      <td>2024-11-24 04:30:00Z</td>
    </tr>
    <tr>
      <td>BP - Diastolic</td>
      <td>89 mm[Hg]</td>
      <td>2024-11-24 04:30:00Z</td>
    </tr>
  </tbody>
</table>
</text>
```



```

</tr>
<tr>
  <td>BP - Systolic</td>
  <td>118 mm[Hg]</td>
  <td>2024-11-24 04:30:00Z</td>
</tr>
<tr>
  <td>hr</td>
  <td>90 /min</td>
  <td>2024-11-24 04:30:00Z</td>
</tr>
<tr>
  <td>rr</td>
  <td>30 /min</td>
  <td>2024-11-24 04:30:00Z</td>
</tr>
</tbody>
</table>
</text>

```

## Coded Entries

This contains the structured data for vital signs. Each `<entry>` corresponds to individual vital sign observations.

```

<entry typeCode="DRIV">
  <organizer classCode="CLUSTER" moodCode="EVN">
    <templateId root="2.16.840.1.113883.10.20.22.4.26" extension="2015-08-01"/>
    <code code="46680005" codeSystem="2.16.840.1.113883.6.96" codeSystemName="SNOMED CT"
displayName="Vital signs">
      <translation code="74728-7" codeSystem="2.16.840.1.113883.6.1" codeSystemName="LOINC"
displayName="Vital signs"/>
    </code>
    <statusCode code="completed"/>
    <effectiveTime>
      <low value="20241124"/>
    </effectiveTime>
    <component>
      <observation classCode="OBS" moodCode="EVN">
        <templateId root="2.16.840.1.113883.10.20.22.4.27" extension="2014-06-09"/>
        <id root="1" extension="vital1"/>
        <code code="59408-5" codeSystem="2.16.840.1.113883.6.1" codeSystemName="LOINC"
displayName="o2_sats"/>
        <statusCode code="completed"/>
        <effectiveTime value="20241124043000+0000"/>
        <value xsi:type="PQ" value="92" unit="%" />
      </observation>
    </component>
  </organizer>
</entry>

```

```

<component>
  <observation classCode="OBS" moodCode="EVN">
    <templateId root="2.16.840.1.113883.10.20.22.4.27" extension="2014-06-09"/>
    <id root="2" extension="vital2"/>
    <code code="8310-5" codeSystem="2.16.840.1.113883.6.1" codeSystemName="LOINC"
displayName="temp"/>
    <statusCode code="completed"/>
    <effectiveTime value="20241124043000+0000"/>
    <value xsi:type="PQ" value="38" unit="Cel"/>
  </observation>
</component>
</organizer>
</entry>

```

## Key elements in Coded Entries

| Element         | Description                                                                                                          |
|-----------------|----------------------------------------------------------------------------------------------------------------------|
| <entry>         | Represents the entire vital sign organizer                                                                           |
| <organizer>     | Groups related observations (e.g., temperature, BP)                                                                  |
| <code>          | Identifies the type of measurement (e.g., vital signs)                                                               |
| <statusCode>    | Indicates the status of the observation (e.g., completed)                                                            |
| <effectiveTime> | Specifies the observation date or time range                                                                         |
| <component>     | Holds each individual observation within the organizer                                                               |
| <observation>   | Captures details of each measured vital sign                                                                         |
| <value>         | Captures the outcome of the assessment (e.g., no impairment) and links to a standardized code (e.g., SNOMED-CT code) |

## History of Encounters

The Encounters section documents the patient's clinical visits or interactions with healthcare providers. It includes encounter details such as the date, time, location, purpose of the visit, and the attending clinician.

### Example XML

The following example sits within a history of encounters `<section></section>` set of tags.

#### Metadata

This defines the section metadata, such as the template ID and LOINC code.

```
<templateId root="2.16.840.1.113883.10.20.22.2.22.1" extension="2015-08-01"/>
<code code="46240-8" codeSystem="2.16.840.1.113883.6.1" codeSystemName="LOINC" displayName="History
of encounters"/>
<title>ENCOUNTERS</title>
```

#### Human-Readable Text

This provides the narrative representation of a patient's history of encounters in a table format.

```
<text>
<table border="1" width="100%">
  <thead>
    <tr>
      <th>Encounter Diagnosis</th>
      <th>Location</th>
      <th>Date</th>
    </tr>
  </thead>
  <tbody>
    <tr>
      <td ID="Enc_7d98b4a4-c68f-4640-8bc7-55362f2cea1a">Fever</td>
      <td>Neighborhood Physicians Clinic</td>
      <td>2015-06-22 10:00 EST
        - 2015-06-22 11:00 EST</td>
    </tr>
  </tbody>
</table>
</text>
```

## Coded Entries

This contains the structured data for a patient's history of encounters. Each `<entry>` corresponds to individual encounter details, such as diagnosis, date, and location.

```
<entry typeCode="DRIV">
  <encounter classCode="ENC" moodCode="EVN">
    <templateId root="2.16.840.1.113883.10.20.22.4.49" extension="2015-08-01"/>
    <id root="1fc3fcd7-f0e5-428a-918e-c00ea3f9a820"/>
    <code code="99202" displayName="Office outpatient visit (new patient)" codeSystemName="CPT-4"
codeSystem="2.16.840.1.113883.6.12">
      <originalText>
        <reference value="#Enc_7d98b4a4-c68f-4640-8bc7-55362f2cea1a"/>
      </originalText>
      <translation code="AMB" codeSystem="2.16.840.1.113883.5.4" displayName="Ambulatory"
codeSystemName="HL7 ActEncounterCode"/>
    </code>
    <effectiveTime value="20241209">
      <low value="20150622100000-0500"/>
      <high value="20150622110000-0500"/>
    </effectiveTime>
    <performer>
      <assignedEntity>
        <id extension="7099467129" root="2.16.840.1.113883.4.6"/>
        <code code="59058001" codeSystem="2.16.840.1.113883.6.96" codeSystemName="SNOMED-CT"
displayName="General Physician"/>
      </assignedEntity>
    </performer>
    <entryRelationship typeCode="SUBJ">
      <act classCode="ACT" moodCode="EVN">
        <templateId root="2.16.840.1.113883.10.20.22.4.80" extension="2015-08-01"/>
        <templateId root="2.16.840.1.113883.10.20.22.4.80"/>
        <templateId root="2.16.840.1.113883.10.20.22.4.19"/>
        <id root="332bf766-75dc-4aa6-99d0-d21db8ad1aff"/>
        <code xsi:type="CE" code="29308-4" codeSystem="2.16.840.1.113883.6.1" codeSystemName="LOINC"
displayName="ENCOUNTER DIAGNOSIS"/>
        <effectiveTime>
          <low value="20150622"/>
        </effectiveTime>
      </act>
      <entryRelationship typeCode="SUBJ" inversionInd="false">
        <observation classCode="OBS" moodCode="EVN">
          <templateId root="2.16.840.1.113883.10.20.22.4.4" extension="2015-08-01"/>
          <templateId root="2.16.840.1.113883.10.20.22.4.4"/>
          <id root="332bf766-75dc-4aa6-99d0-d21db8ad1aff"/>
          <code code="404684003" displayName="Finding" codeSystemName="SNOMED-CT"
codeSystem="2.16.840.1.113883.6.96">
            <translation code="75321-0" codeSystem="2.16.840.1.113883.6.1" codeSystemName="LOINC"
displayName="Clinical finding"/>
          </code>
```

```

    <statusCode code="completed"/>
    <effectiveTime>
      <low value="20150622"/>
    </effectiveTime>
    <value displayName="Fever" code="386661006" xsi:type="CD" codeSystem="2.16.840.1.113883.6.96"
codeSystemName="SNOMED CT"/>
  </observation>
</entryRelationship>
</act>
</entryRelationship>
</encounter>
</entry>

```

## Key elements in Coded Entries

| Element             | Description                                                                      |
|---------------------|----------------------------------------------------------------------------------|
| <entry>             | Represents the structured data for an encounter                                  |
| <encounter>         | Holds details of the clinical encounter                                          |
| <templateId>        | Identifies the CDA template being used                                           |
| <code>              | Identifies the type of data being documented (e.g., diagnosis code)              |
| <effectiveTime>     | Specifies the date or time range of the encounter                                |
| <performer>         | Identifies the clinician or entity responsible for the encounter or observation. |
| <entryRelationship> | Establishes relationships to nested clinical observations                        |
| <observation>       | Captures the diagnosis or findings related to the encounter                      |
| <value>             | Contains the coded diagnosis or clinical finding                                 |
| <statusCode>        | Indicates the status of the observation (e.g., completed)                        |

## History of Immunizations

This section contains the patient's vaccination history, including immunization types, dates administered, and, if applicable, notes on vaccine status or contraindications.

### Example XML

The following example sits within a history of immunizations `<section></section>` set of tags.

#### Metadata

This defines the section metadata, such as the template ID and LOINC code.

```
<templateId root="2.16.840.1.113883.10.20.22.2.2.1" extension="2014-06-09"/>
<code code="11369-6" codeSystem="2.16.840.1.113883.6.1" codeSystemName="LOINC" displayName="History
of immunizations"/>
<title>IMMUNIZATIONS</title>
```

#### Human-Readable Text

This provides the narrative representation of a patient's history of immunizations in a table format.

```
<text>
<table border="1" width="100%">
  <thead>
    <tr>
      <th>Vaccine</th>
      <th>Date</th>
      <th>Status</th>
    </tr>
  </thead>
  <tbody>
    <tr>
      <td>Tetanus and diphtheria toxoids</td>
      <td>2015-06-22</td>
      <td>Completed</td>
    </tr>
  </tbody>
</table>
</text>
```

## Coded Entries

This contains the structured data for a patient's history of immunizations. Each `<entry>` documents immunization details, including the vaccine administered, its status, and the date.

```
<entry typeCode="DRIV">
  <substanceAdministration classCode="SBADM" moodCode="EVN" negationInd="false">
    <templateId root="2.16.840.1.113883.10.20.22.4.52" extension="2015-08-01"/>
    <id root="15cc940a-e88a-45f6-b454-720be841f489"/>
    <statusCode code="completed"/>
    <effectiveTime value="20120104"/>
    <consumable>
      <manufacturedProduct classCode="MANU">
        <templateId root="2.16.840.1.113883.10.20.22.4.54" extension="2014-06-09"/>
        <manufacturedMaterial>
          <code displayName="Tetanus and diphtheria toxoids"
            code="106" codeSystem="2.16.840.1.113883.12.292" codeSystemName="CVX"/>
        </manufacturedMaterial>
      </manufacturedProduct>
    </consumable>
  </substanceAdministration>
</entry>
```

## Key elements in Coded Entries

| Element                                      | Description                                                                                       |
|----------------------------------------------|---------------------------------------------------------------------------------------------------|
| <code>&lt;entry&gt;</code>                   | Wraps the immunization record and identifies it as a structured data entry                        |
| <code>&lt;substanceAdministration&gt;</code> | Describes the administration of a vaccine, including the details of what was given and its status |
| <code>&lt;statusCode&gt;</code>              | Represents the current status of the immunization (e.g., "completed," "canceled")                 |
| <code>&lt;effectiveTime&gt;</code>           | Specifies the date or time the immunization was administered                                      |
| <code>&lt;consumable&gt;</code>              | Contains details about the vaccine or substance administered                                      |
| <code>&lt;manufacturedMaterial&gt;</code>    | Provides the name and standard code for the vaccine                                               |
| <code>&lt;code&gt;</code>                    | The code value for the vaccine (e.g., "149" for Influenza LAIV4)                                  |

## Care Team

The Care Team Section documents the members of the care team involved in a patient's care, including their roles, names, and contact details.

### Example XML

The following example sits within a care team `<section></section>` set of tags.

#### Metadata

This defines the section metadata, such as the template ID and LOINC code.

```
<templateId root="2.16.840.1.113883.10.20.22.2.500" extension="2019-07-01"/>
<code code="85847-2" codeSystem="2.16.840.1.113883.6.1" codeSystemName="LOINC" displayName="Care
Team Section"/>
<title>Care Team Section</title>
```

#### Human-Readable Text

This provides the narrative representation of a patient's care team in a table format.

```
<text>
<table border="1" width="100%">
  <thead>
    <tr>
      <th>Performer Name</th>
      <th>Performer Role</th>
      <th>Since</th>
    </tr>
  </thead>
  <tbody>
    <tr>
      <td>Albert Davis</td>
      <td>Care team member</td>
      <td>2015-06-22</td>
    </tr>
  </tbody>
</table>
</text>
```



## Coded Entries

This contains the structured data for a patient's care team. Each `<entry>` corresponds to a specific care team member, with structured data detailing their role, name, and organization.

```
<entry>
  <organizer classCode="CLUSTER" moodCode="EVN">
    <templateId root="2.16.840.1.113883.10.20.22.4.500" extension="2019-07-01"/>
    <id root="1fc3fcd7-f0e5-428a-918e-c00ea3f9a820"/>
    <code code="86744-0" codeSystem="2.16.840.1.113883.6.1" codeSystemName="LOINC" displayName="Care
Team"/>
    <statusCode code="active"/>
    <effectiveTime>
      <low value="20150622150000+0000"/>
    </effectiveTime>
    <component>
      <act classCode="PCPR" moodCode="EVN">
        <templateId root="2.16.840.1.113883.10.20.22.4.500.1" extension="2019-07-01"/>
        <id root="97f3b7a4-8126-400a-9316-ae79209f9a76"/>
        <code code="85847-2" codeSystem="2.16.840.1.113883.6.1" codeSystemName="LOINC"
displayName="Care Team Section"/>
        <statusCode code="active"/>
        <effectiveTime>
          <low value="20150622100000+0000"/>
        </effectiveTime>
        <performer typeCode="PRF">
          <assignedEntity>
            <id root="2.16.840.1.113883.4.6" extension="1234567890"/>
            <code displayName="Doctor" code="200000000X" codeSystem="2.16.840.1.113883.6.101"
codeSystemName="NUCC Health Care Provider Taxonomy"/>
            <addr>
              <streetAddressLine>2472 Rocky Place</streetAddressLine>
              <city>Beaverton</city><state>OR</state>
              <postalCode>97006</postalCode>
              <country>US</country>
            </addr>
            <telecom use="WP" value="tel:+15555551002"/>
            <assignedPerson>
              <name>
                <given>Albert</given>
                <family>Davis</family>
              </name>
            </assignedPerson>
          </assignedEntity>
        </performer>
      </act>
    </component>
  </organizer>
</entry>
```

## Key elements in Coded Entries

| Element          | Description                                                                                  |
|------------------|----------------------------------------------------------------------------------------------|
| <entry>          | Wraps the care team information as a structured entry                                        |
| <organizer>      | Groups related information about the care team                                               |
| <statusCode>     | Indicates the status of the care team (e.g., "active")                                       |
| <effectiveTime>  | Specifies the time period for which the care team member's involvement is valid              |
| <performer>      | Details the care team member, including their role, contact information, and assigned entity |
| <assignedEntity> | Captures the care team member's organizational and role details                              |
| <assignedPerson> | Provides the name of the care team member                                                    |
| <code>           | Specifies the role of the care team member (e.g., "Doctor")                                  |
| <addr>           | Provides the care team member's address                                                      |
| <telecom>        | Provides the care team member's contact details                                              |

## Assessments

The Assessments Section contains clinical evaluations or assessments of a patient's condition, which are typically part of a summary note or care plan.

### Example XML

The following example sits within an assessment `<section></section>` set of tags.

#### Metadata

This defines the section metadata, such as the template ID and LOINC code.

```
<templateId root="2.16.840.1.113883.10.20.22.2.8"/>
<code codeSystem="2.16.840.1.113883.6.1" codeSystemName="LOINC" code="51848-0"
displayName="ASSESSMENTS"/>
<title>ASSESSMENTS</title>
```

#### Human-Readable Text

This provides the narrative representation of a patient's assessments in a table format.

```
<text>
  Fever - The patient was found to have fever and Dr Davis is suspecting Anemia based on the patient history.
  So Dr Davis asked the patient to closely monitor the temperature and blood pressure and get admitted to
  Community Health Hospitals if the fever does not subside within a day.
</text>
```

#### Coded Entries

This contains the structured data for a patient's assessments. Each `<entry>` represents a structured clinical assessment or diagnosis with relevant metadata and codes.

```
<entry>
  <act classCode="ACT" moodCode="EVN">
    <templateId root="2.16.840.1.113883.10.20.22.4.202" extension="2016-11-01"/>
    <id root="12345678-aaaa-1234-bbbb-987654321"/>
    <code code="51848-0" codeSystem="2.16.840.1.113883.6.1" codeSystemName="LOINC"
displayName="ASSESSMENTS"/>
    <statusCode code="completed"/>
    <effectiveTime>
      <low value="20241209"/>
    </effectiveTime>
  </act>
</entry>
```

```

</effectiveTime>
<entryRelationship typeCode="SUBJ">
  <observation classCode="OBS" moodCode="EVN">
    <templateId root="2.16.840.1.113883.10.20.22.4.4"/>
    <id root="abcd1234-5678-efgh-ijkl-1234567890"/>
    <code code="404684003" displayName="Finding" codeSystem="2.16.840.1.113883.6.96"
codeSystemName="SNOMED CT"/>
    <statusCode code="completed"/>
    <value xsi:type="CD" code="285381006" displayName="Acute infective exacerbation of chronic
obstructive airways disease" codeSystem="2.16.840.1.113883.6.96" codeSystemName="SNOMED CT"/>
  </observation>
</entryRelationship>
</act>
</entry>

```

## Key elements in Coded Entries

| Element                     | Description                                                                    |
|-----------------------------|--------------------------------------------------------------------------------|
| <entry>                     | Wraps the structured assessment data                                           |
| <act>                       | Represents the clinical activity (e.g., an assessment)                         |
| <statusCode>                | Indicates the status of the assessment (e.g., "completed")                     |
| <effectiveTime>             | Represents the time period when the assessment was valid                       |
| <entryRelationship>         | Links the act to an observation, typically for a specific finding or diagnosis |
| <observation>               | Contains the detailed assessment data                                          |
| <code> (within observation) | Represents the SNOMED-CT code for the clinical finding (e.g., "Finding")       |
| <value>                     | The coded value of the assessment (e.g., specific diagnosis or condition)      |

## Treatment Plan

The Treatment Plan Section contains information regarding planned care, treatments, or procedures intended to address a patient's health conditions. This section typically includes entries for scheduled or recommended care.

### Example XML

The following example sits within a treatment plan `<section></section>` set of tags.

#### Metadata

This defines the section metadata, such as the template ID and LOINC code.

```
<templateId root="2.16.840.1.113883.10.20.22.2.10" extension="2014-06-09"/>
<code code="18776-5" codeSystem="2.16.840.1.113883.6.1" codeSystemName="LOINC"
displayName="Treatment Plan"/>
<title>TREATMENT PLAN</title>
```

#### Human-Readable Text

This provides the narrative representation of a treatment plan in a table format.

```
<text>
<table border="1" width="100%">
  <thead>
    <tr>
      <th>Planned Care</th>
    </tr>
  </thead>
  <tbody>
    <tr>
      <td ID="Plan_78c442e0-c37a-4caf-9fa1-62b7e02727e1__0">Assistance with application for food pantry
program</td>
    </tr>
  </tbody>
</table>
</text>
```

## Coded Entries

This contains the structured data for the treatment plan. Each `<entry>` represents a structured plan of care, including any scheduled treatments or recommendations.

```
<entry>
  <procedure moodCode="RQO" classCode="PROC">
    <templateId root="2.16.840.1.113883.10.20.22.4.41" extension="2022-06-01"/>
    <id root="12345678-abcd-1234-efgh-9876543210"/>
    <code code="467771000124109" codeSystem="2.16.840.1.113883.6.96" codeSystemName="SNOMED CT"
      displayName="Assistance with application for food pantry program"/>
    <text>
      <reference value="Plan_78c442e0-c37a-4caf-9fa1-62b7e02727e1__0"/>
    </text>
    <statusCode code="active"/>
    <effectiveTime value="20241201"/>
  </procedure>
</entry>
```

## Key elements in Coded Entries

| Element             | Description                                                                                      |
|---------------------|--------------------------------------------------------------------------------------------------|
| <entry>             | Wraps the structured treatment plan data                                                         |
| <procedure>         | Represents the planned clinical action or service                                                |
| <templateId>        | Identifies the CCDA Plan of Care Activity Procedure template                                     |
| <statusCode>        | Indicates the status of the intervention (e.g., "active" or "planned")                           |
| <effectiveTime>     | Represents the time when the planned treatment is scheduled to begin or applies                  |
| <entryRelationship> | Links the treatment act to a specific observation (e.g., planned procedure details)              |
| <observation>       | Contains details of the planned care, such as a specific procedure or therapy                    |
| <code>              | SNOMED CT code describing the planned intervention                                               |
| <text>              | Human-readable reference linking the structured entry back to the narrative Plan of Care section |

## Clinical Notes

The Clinical Notes Section provides detailed narrative summaries or observations documented during a patient's care. These notes can include various types of clinical documentation such as progress notes, emergency department notes, discharge summaries, and more.

### Example XML

The following example sits within the clinical notes `<section></section>` set of tags.

#### Metadata

This defines the section metadata, such as the template ID and LOINC code.

```
<templateId root="2.16.840.1.113883.10.20.22.2.65" extension="2016-11-01"/>
<code code="11506-3" codeSystem="2.16.840.1.113883.6.1" codeSystemName="LOINC" displayName="Clinical
Notes Section"/>
<title>Clinical Notes Section</title>
```

#### Human-Readable Text

This provides the narrative representation of the clinical notes in a table format.

```
<text>
<table border="1" width="100%">
  <thead>
    <tr>
      <th>Type</th>
      <th>Summary</th>
      <th>Written by:</th>
      <th>Date</th>
    </tr>
  </thead>
  <tbody>
    <tr>
      <td>Progress Note</td>
      <td ID="Note_38751ca6-d1dd-4a33-80e8-ec65346b7867">
        Ms Alice Newman seems to be developing high fever and hence we recommend her to get admitted to a
        nearby hospital using the provided referral.
      </td>
      <td>Albert Davis MD</td>
      <td>2015-06-22 10:00 EST</td>
    </tr>
  </tbody>
</table>
</text>
```



## Coded Entries

This contains the structured data for the clinical notes. Each `<entry>` contains structured data corresponding to a specific clinical note.

```
<entry>
  <act classCode="ACT" moodCode="EVN">
    <templateId root="2.16.840.1.113883.10.20.22.4.202" extension="2016-11-01"/>
    <code code="34109-9" codeSystem="2.16.840.1.113883.6.1" codeSystemName="LOINC"
displayName="Note">
      <translation code="11506-3" codeSystem="2.16.840.1.113883.6.1" codeSystemName="LOINC"
displayName="Progress Note"/>
    </code>
    <text>
      <reference value="#Note_38751ca6-d1dd-4a33-80e8-ec65346b7867"/>
    </text>
    <statusCode code="completed"/>
    <effectiveTime value="20150622"/>
    <author>
      <templateId root="2.16.840.1.113883.10.20.22.4.119"/>
      <time value="20150622100000-0500"/>
      <assignedAuthor>
        <id root="2.16.840.1.113883.4.6" extension="7099467129"/>
        <code code="200000000X" displayName="Doctor" codeSystem="2.16.840.1.113883.6.101"
codeSystemName="NUCC Health Care Provider Taxonomy"/>
        <assignedPerson>
          <name>
            <given>Albert</given>
            <family>Davis</family>
          </name>
        </assignedPerson>
        <representedOrganization>
          <name>Neighborhood Physicians Practice</name>
        </representedOrganization>
      </assignedAuthor>
    </author>
  </act>
</entry>
```

## Key elements in Coded Entries

| Element                   | Description                                                           |
|---------------------------|-----------------------------------------------------------------------|
| <entry>                   | Wraps the structured data for the clinical note                       |
| <act>                     | Represents the clinical note activity                                 |
| <text>                    | Provides a reference link to the human-readable text section          |
| <statusCode>              | Indicates the status of the note (e.g., "completed")                  |
| <effectiveTime>           | Specifies the date and time when the note was documented              |
| <author>                  | Captures the healthcare provider who wrote the note                   |
| <assignedAuthor>          | Contains the author's credentials, such as ID, role, and organization |
| <assignedPerson>          | Holds the author's name                                               |
| <representedOrganization> | Identifies the organization the author represents                     |

## Payers

The Health Insurance Information section lists any data related to an individual's insurance coverage for healthcare. Insurance information includes coverage status, coverage type, relationship to subscriber, payer identifier, as well as member, subscriber, and group identifiers.

### Example XML

The following example sits within a health insurance information `<section></section>` set of tags.

#### Metadata

This defines the section metadata, such as the template ID and LOINC code.

```
<templateId root="2.16.840.1.113883.10.20.22.2.18" extension="2015-08-01"/>
<code code="48768-6" codeSystem="2.16.840.1.113883.6.1" codeSystemName="LOINC"
displayName="Payers"/>
<title>Insurance Providers / Payers</title>
```

#### Human-Readable Text

This provides the narrative representation of health insurance information in a table format.

```
<text>
  <table border="1" width="100%">
    <thead>
      <tr>
        <th>Payer</th>
        <th>Plan name</th>
        <th>Member ID</th>
        <th>Relationship</th>
        <th>Effective date</th>
      </tr>
    </thead>
    <tbody>
      <tr>
        <td>Example Payer Organization</td>
        <td>Acme Gold Plan</td>
        <td>88800933502</td>
        <td>Family Dependent</td>
        <td>2015-01-01</td>
      </tr>
    </tbody>
  </table>
</text>
```

## Coded Entries

This contains the structured data for a patient's health information. Each `<entry>` corresponds to individual payer details.

```
<entry typeCode="DRIV">
  <act classCode="ACT" moodCode="EVN">
    <templateId root="2.16.840.1.113883.10.20.22.4.60" extension="2015-08-01"/>
    <id root="dcf725d7-e4d7-47bb-9618-7f8b6e39b583"/>
    <code code="48768-6" codeSystem="2.16.840.1.113883.6.1" codeSystemName="LOINC"
displayName="Payment sources"/>
    <statusCode code="completed"/>
    <effectiveTime value="20150101"/>
    <entryRelationship typeCode="COMP">
      <act classCode="ACT" moodCode="EVN">
        <templateId root="2.16.840.1.113883.10.20.22.4.61"/>
        <templateId root="2.16.840.1.113883.10.20.22.4.61" extension="2015-08-01"/>
        <id root="dcf725d7-e4d7-47bb-9618-7f8b6e39b583" extension="Acme Gold Plan"/>
        <code code="72" displayName="PPO" codeSystem="2.16.840.1.113883.3.221.5"
codeSystemName="Source of Payment Typology (PHDSC)"/>
        <statusCode code="completed"/>
        <performer typeCode="PRF">
          <templateId root="2.16.840.1.113883.10.20.22.4.87"/>
          <time>
            <low nullFlavor="UNK"/>
            <high nullFlavor="UNK"/>
          </time>
          <assignedEntity>
            <id extension="11223344" root="2.16.840.1.113883.6.300"/>
            <code code="PAYOR" codeSystem="2.16.840.1.113883.5.110" codeSystemName="HL7
RoleCode"/>
            <addr use="WP">
              <streetAddressLine>1234 Insurance Road</streetAddressLine>
              <city>Blue Bell</city>
              <state>MA</state>
              <postalCode>02368</postalCode>
              <country>US</country>
            </addr>
            <representedOrganization>
              <name>Example Payer Organization</name>
            </representedOrganization>
            <addr use="WP">
              <streetAddressLine>1234 Insurance Road</streetAddressLine>
              <city>Blue Bell</city>
              <state>MA</state>
              <postalCode>02368</postalCode>
```

```

    <country>US</country>
  </addr>
</representedOrganization>
</assignedEntity>
</performer>
<entryRelationship typeCode="REFR">
  <act classCode="ACT" moodCode="DEF">
    <id root="dcf725d7-e4d7-47bb-9618-7f8b6e39b583"/>
    <code nullFlavor="NI"/>
    <text>5553652</text>
  </act>
</entryRelationship>
</act>
</entryRelationship>
</act>
</entry>

```

## Key elements in Coded Entries

| Element                          | Description                                                                                                         |
|----------------------------------|---------------------------------------------------------------------------------------------------------------------|
| <entry>                          | Wraps the structured insurance / payment source information for the patient                                         |
| <act>                            | Represents the Payment Sources section container, using LOINC to indicate the type of financial data being captured |
| <templateId>                     | Identifies the CCDA Payment Sources or Coverage Activity template                                                   |
| <code>                           | Specifies that this entry is payment source information                                                             |
| <statusCode>                     | Indicates the status of the payment source documentation                                                            |
| <effectiveTime>                  | Date when coverage became effective                                                                                 |
| <author>                         | Captures the healthcare provider who wrote the note                                                                 |
| <act> (within Coverage Activity) | Identifies the specific coverage activity                                                                           |
| <performer>                      | Payer entity responsible for coverage                                                                               |
| <representedOrganization>        | Identifies the organization providing coverage                                                                      |
| <text>                           | Contains specific subscriber or policy ID                                                                           |

## Participant

The participant section lists any persons with a legal or familial relationship to the patient. Each entry includes the name of the participant and the relationship of the participant to the patient.

### Example XML

The following example sits within a `<participant></participant>` set of tags.

#### Metadata

This defines the section metadata, such as the template ID.

```
<templateId root="2.16.840.1.113883.10.20.22.5.8" extension="2023-05-01"/>
```

#### Human-Readable Text

N/A

#### Coded Entries

This contains the structured data for a patient's related persons. Each `<participant>`, and within each `<participant>`, each `<associatedEntity>` corresponds to an individual participant.

```
<participant typeCode="IND">
  <templateId root="2.16.840.1.113883.10.20.22.5.8"/>
  <templateId root="2.16.840.1.113883.10.20.22.5.8" extension="2023-05-01"/>
  <associatedEntity classCode="PRS">
    <code code="GRPRN" displayName="Grandparent" codeSystem="2.16.840.1.113883.1.11.19563"
codeSystemName="Personal Relationship Role Type Value Set"/>
    <addr>
      <streetAddressLine>1357 Amber Dr.</streetAddressLine>
      <city>Beaverton</city>
      <state>OR</state>
      <postalCode>97006</postalCode>
      <country>USA</country>
    </addr>
    <associatedPerson>
      <name>
        <given>Rick</given>
        <family>Holler</family>
      </name>
    </associatedPerson>
  </associatedEntity>
```

</participant>

## Key elements in Coded Entries

| Element            | Description                                                             |
|--------------------|-------------------------------------------------------------------------|
| <participant>      | Represents the structured data for a participant                        |
| <templateId>       | Identifies the CDA template being used                                  |
| <code>             | Describes the personal relationship between the participant and patient |
| <addr>             | Specifies the address of the participant                                |
| <associatedPerson> | Identifies the participant that is related to the patient               |

## Supplemental Data

For each patient, the EHI export includes comma separated values (csv) files for any additional electronic health information not captured in the patient's C-CDA. We categorize this supplemental data into three categories: documents, orders, and charges. These csv files contain well formatted json columns.

### Documents

| Field                           | Description                                    | Field Data Type       |
|---------------------------------|------------------------------------------------|-----------------------|
| <code>class</code>              | The class of document                          | <code>string</code>   |
| <code>type</code>               | The type of document                           | <code>string</code>   |
| <code>awaiting_signature</code> | Signature status of the document               | <code>boolean</code>  |
| <code>timestamp</code>          | The date and time associated with the document | <code>datetime</code> |
| <code>added_at</code>           | The date and time the document was added       | <code>datetime</code> |
| <code>source</code>             | The source of document                         | <code>string</code>   |
| <code>desc</code>               | The description of document                    | <code>string</code>   |
| <code>loinc_id</code>           | The LOINC code of document                     | <code>string</code>   |
| <code>text_data</code>          | Text data associated with the document         | <code>string</code>   |
| <code>json_data</code>          | JSON data associated with the document         | <code>dict</code>     |
| <code>application_data</code>   | Application data associated with the document  | <code>dict</code>     |
| <code>visit</code>              | The visits associated with the document        | <code>uuid</code>     |



## Orders

| Field                        | Description                                            | Field Data Type       |
|------------------------------|--------------------------------------------------------|-----------------------|
| <code>name</code>            | The name of the order                                  | <code>string</code>   |
| <code>label</code>           | The label of the order                                 | <code>string</code>   |
| <code>serial_number</code>   | Serial number of the order                             | <code>int</code>      |
| <code>data</code>            | Data associated with the order                         | <code>dict</code>     |
| <code>status</code>          | The status of the order                                | <code>string</code>   |
| <code>urgency</code>         | The urgency of the order                               | <code>string</code>   |
| <code>reason</code>          | The reason of the order                                | <code>string</code>   |
| <code>start_time</code>      | The start date and time of the order                   | <code>datetime</code> |
| <code>end_time</code>        | The end date and time of the order                     | <code>datetime</code> |
| <code>details</code>         | The details of the order                               | <code>string</code>   |
| <code>fully_signed_at</code> | The date and time the order was signed                 | <code>datetime</code> |
| <code>ixp_start_time</code>  | The investigation start time associated with the order | <code>datetime</code> |
| <code>ixp_end_time</code>    | The investigation end time associated with the order   | <code>datetime</code> |
| <code>visit</code>           | The visit associated with the order                    | <code>uuid</code>     |

## Charges

| Field          | Description                                          | Field Data Type |
|----------------|------------------------------------------------------|-----------------|
| rate           | The rate of the charge                               | float           |
| revenue_code   | The revenue code of the charge                       | string          |
| procedure_code | The procedure code of the charge                     | string          |
| modifier       | The modifier associated with the charge              | string          |
| pos_code       | The point of service code associated with the charge | string          |
| dept           | The department associated with the charge            | string          |
| quantity       | The quantity of the charge                           | float           |
| total          | The total price of the charge                        | float           |
| recorded_at    | The date and time recorded for the charge            | datetime        |
| service_date   | The service date of the charge                       | date            |
| is_emg         | The emergency status associated with the charge      | boolean         |
| is_epsdt       | The CMS epsdt status associated with the charge      | boolean         |
| is_verified    | The verified status associated with the charge       | boolean         |
| ndc            | The ndc associated with the charge                   | string          |
| deleted_at     | The date and time the charge was deleted             | datetime        |
| diagnosis_ids  | The diagnosis IDs associated with the charge         | array           |